

Opta — 50 mA CT AC MCSA Wiring

MooreVIEW integrator reference

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True line MCSA — burden shunt + mid-rail bias — `MV_CT_WAVEFORM=1`

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Applies to: Arduino Opta MQTT ST firmware (`firmware/arduino-opta-mqtt-st/`)

Path: True line MCSA — `-DMV_CT_WAVEFORM=1` + M4 coprocessor

Purpose

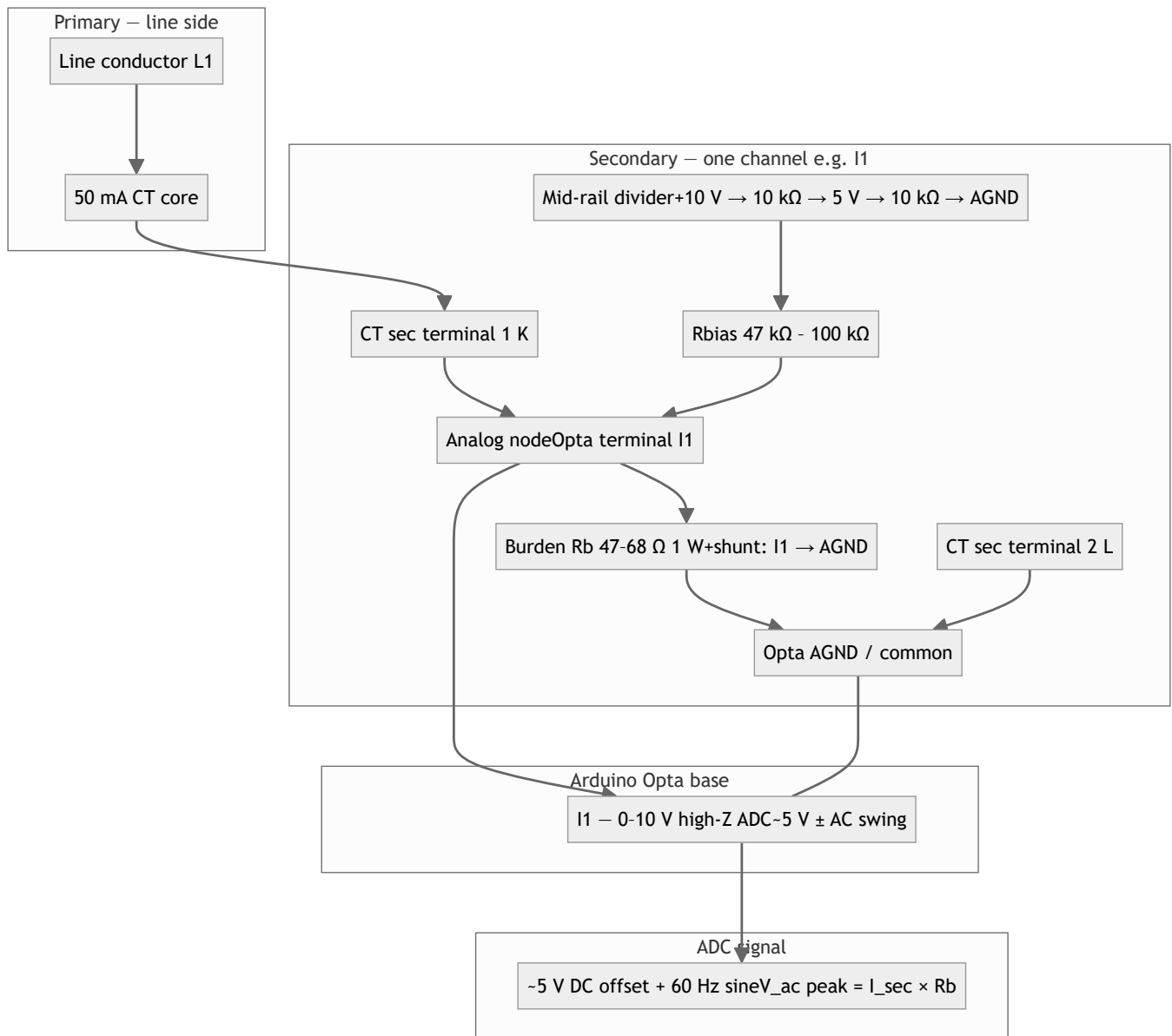
This document shows how to wire **50 mA AC current transformers** to Opta base analog inputs **I1–I6** for **true AC waveform MCSA** (real 60 Hz FFT on the Cortex-M4 coprocessor), as opposed to:

Input type	Opta sees	MCSA mode
0–1 V / 0–10 V DC-RMS transmitter	Steady DC $\propto$ RMS amps	Pseudo MCSA (<code>MV_CT_WAVEFORM=0</code>)
50 mA CT + burden + mid-rail bias	60 Hz sine centered ~5 V	True MCSA (<code>MV_CT_WAVEFORM=1</code>)

Firmware reference: `firmware/arduino-opta-mqtt-st/README.md` — *AC line MCSA* section.

Single channel — AC waveform MCSA

One CT channel (example: Pump 1 phase A → **I1**):

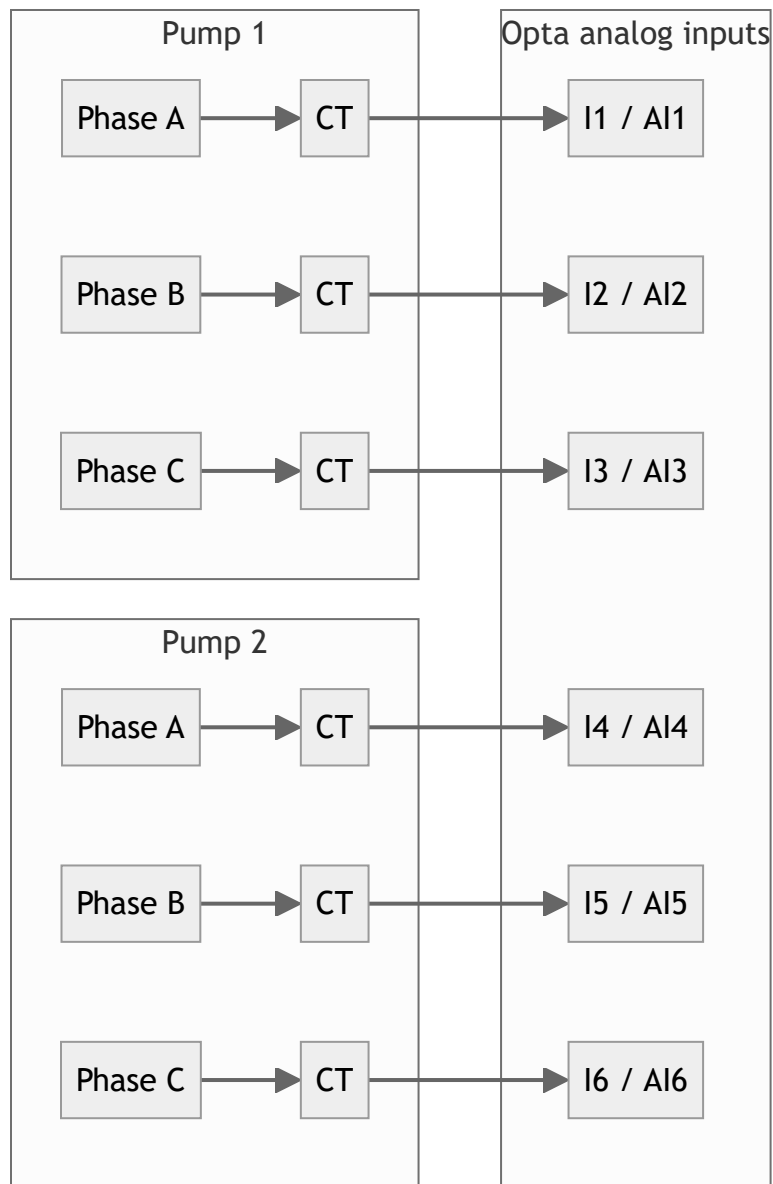


Reading the diagram

- CT secondary is a **current source**; the burden is the **load from I1 to AGND** ($V = I \times R_b$).
- **Do not** place a resistor in **series** between a DC transmitter and I1 — Opta expects **voltage at the pin relative to AGND** (high-Z input).
- Mid-rail bias lifts the idle point to **~5 V** so the sine stays inside **0–10 V**.
- At **50 mA secondary** $\times$ **68 Ω** $\approx$ **3.4 V** peak $\rightarrow$ roughly **5 V $\pm$ 3.4 V** (1.6 V to 8.4 V) — OK.

Duplex lift — six CT channels

Repeat the single-channel circuit for **I1 through I6**:

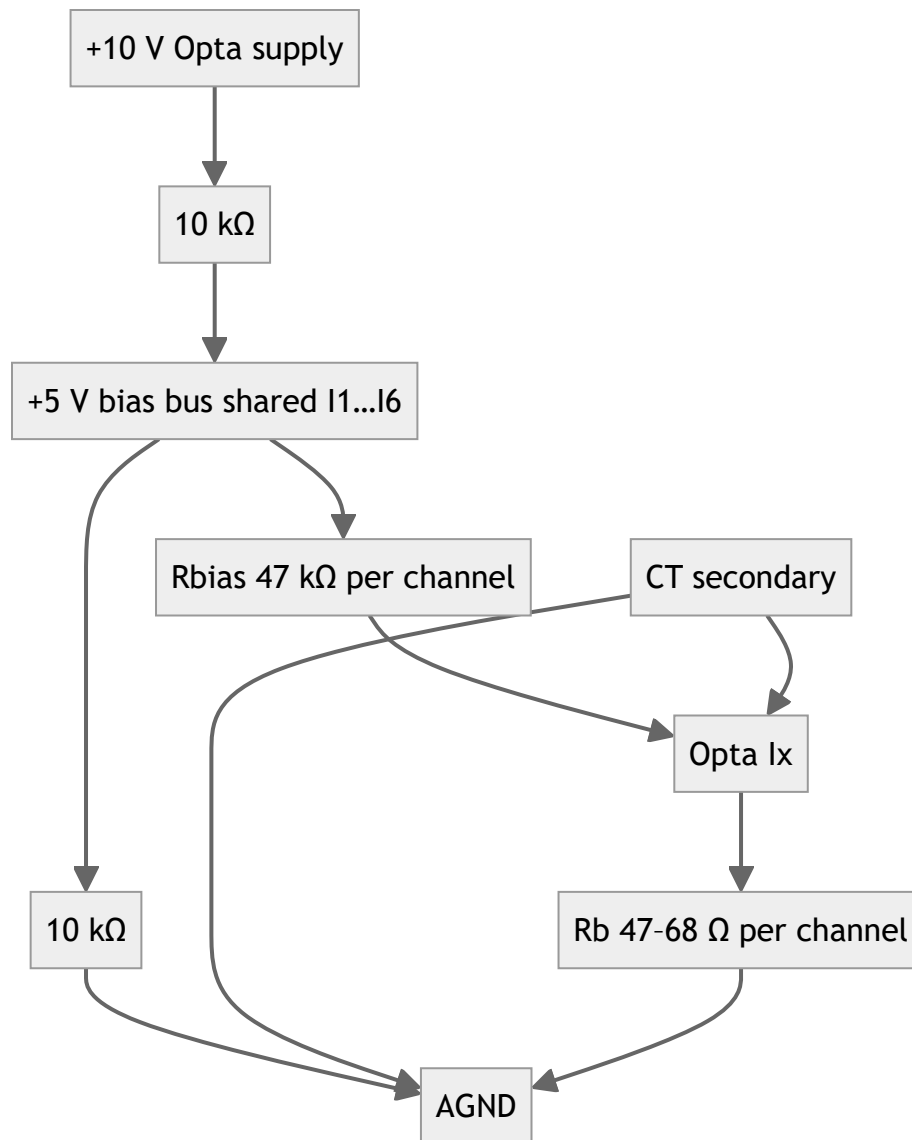


Opta pin	Tag	Pump / phase
I1	AI1	Pump 1 — phase A
I2	AI2	Pump 1 — phase B
I3	AI3	Pump 1 — phase C
I4	AI4	Pump 2 — phase A
I5	AI5	Pump 2 — phase B
I6	AI6	Pump 2 — phase C

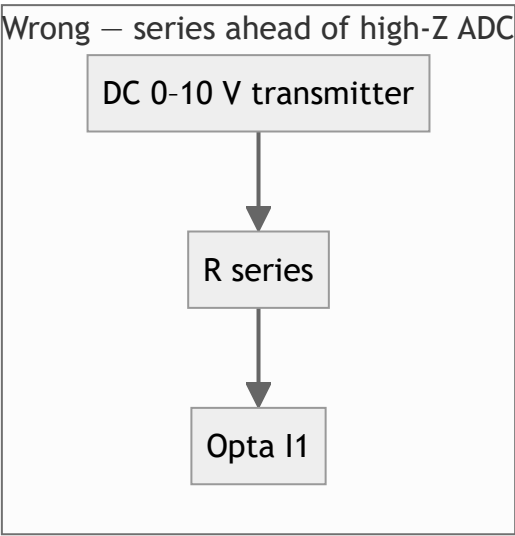
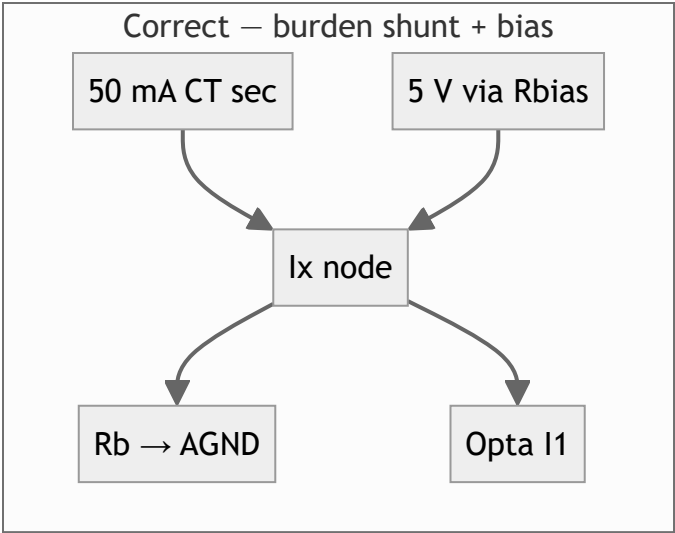
Each channel: **CT sec** → **Ix node** → **Rb (Ix→AGND)** → **Rbias from 5 V bus** → **CT return** → **AGND**.

Bias + burden detail

Shared **+5 V bias bus** can feed all six channels (one divider, six Rbias resistors):



Wrong vs right — “shunt to GND, not series”



Firmware note (`mv_config.h`): *burden/shunt ohms from analog input to GND (not series).*

Burden resistor selection

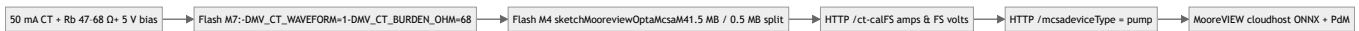
Goal	Burden	At 50 mA sec	Firmware
AC waveform MCSA	47–68 Ω	~2.4–3.4 V peak AC	<code>-DMV_CT_WAVEFORM=1</code>
DC-RMS shunt (pseudo MCSA)	180 Ω	~9 V DC at FS	default <code>MV_CT_WAVEFORM=0</code>

Goal	Burden	At 50 mA sec	Firmware
DC-RMS shunt	100 Ω	~5 V DC at FS	-DMV_CT_BURDEN_OHM=100
DC-RMS shunt	20 Ω	~1 V DC at FS	default (same as 0–1 V TX)

For AC MCSA use **47–68 Ω** so the biased sine does **not clip** at 0 V or 10 V.

Do not use AFX00007 **current** input mode with 50 mA (module max **25 mA**).

Commissioning flow



Pre-power checks

Check	Expected
Motor OFF , DMM on I1	~5 V DC (mid-rail)
Motor RUN , scope on I1	60 Hz sine around 5 V
Peak voltage	0.2 V – 9.8 V (no clipping)
CT secondary	Never open while primary energized — burden always connected

Runtime telemetry

With AC path active, MQTT Parc `runtime` should show:

- `ctWaveform`: "ac"
- `mcsaLite`: false (after M4 ingest)
- `trueFft` / cooked `mcsa[]` with real fund / rotor / bearing bins

Related docs

Document	Topic
firmware/arduino-opta-mqtt-st/README.md	Dual-core MCSA, compile flags
docs/AI_EDGE_CLOUD.md	Pseudo vs true MCSA, host ONNX
docs/projects/LIFT-STATION-EZMETER-PARC-CLOUD.md	Duplex I1–I6 tag map (DC transmitter baseline)

Document	Topic
docs/projects/UNO-Q-EDGE-MOTOR-FAULT.md	UNO Q true FFT alternative

Generate PDF: `npm run build:opta-ct-mcsa-wiring-pdf`